



Cambridge (CIE) IGCSE Computer Science



Your notes

Artificial Intelligence

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Artificial Intelligence (AI)

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What is artificial intelligence?

- Artificial intelligence (**AI**) is a machine that can **simulate intelligent behaviours** similar to that of a **human**
- AI is a system that can:
 - Learn** – acquire new information
 - Decide** – analyse and make choices
 - Act autonomously** – take actions without human input
- There are two main types of AI:
 - Weak AI**, also known as **narrow AI**, is designed to perform a specific task or set of tasks
 - Strong AI**, also known as artificial general intelligence (**AGI**), is designed to perform any intellectual task that a human can do

Advantages and disadvantages of AI

Advantages	Disadvantages
Increased efficiency	Job losses
Increased accuracy	Potential for biased decision making
Scalability	Ethical concerns over its use

Characteristics of AI

- AI shares **three** common characteristics:
 - Collection of data**
 - Rules for using data**
 - Ability to reason**



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Collection of data	Rules for using data	Ability to reason
AI systems require large amounts of data to perform tasks The data is processed using rules or algorithms that enable the system to make decisions and predictions	AI systems can use logical reasoning to evaluate information and make decisions based on that information	It can change its own rules and data

Expert systems

What is an expert system?

- An expert system **mimics human knowledge and experience** and uses a combination of the two to **solve problems or answer questions**
- Examples of expert systems include:
 - Equipment troubleshooting and repair**
 - Technical support**
 - Medical diagnosis system**
- Expert systems share **four key characteristics**

Knowledge base	Rule base	Inference engine	Interface
A database of facts to generate rules that are used to solve problems and make decisions	A set of rules or logic that is used to apply the knowledge in the knowledge base to solve specific problems	A program that applies the rules in the rule base to the facts in the knowledge base to solve problems	A way for users to interact with the system and provide input

Advantages and disadvantages of expert systems

Advantages	Disadvantages
Consistent results	Only as good as the data entered



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Faster responses	Responses can be cold and lack human emotion
Can store large amounts of data	Requires training to use correctly
Unbiased	

Machine Learning

What is machine learning?

- Machine learning is **one method that can help** to achieve an artificial intelligence (AI)
- By giving a machine **data** so that it can '**learn over time**' it helps towards **training** a machine or software to **perform a task** and **improve** its accuracy and efficiency
- Uses **algorithms** to analyse data and **identify patterns** or relationships

Advantages	Disadvantages
Reduces the need for manual work, saving time and effort	Needs vast amounts of quality data to perform well.
Can detect patterns and make predictions more accurately than humans in many cases	Requires high processing power and resources.
Continuously improves performance as it processes more data	



Worked Example

Describe the characteristics of AI

[3]

Answer

Any **three** from:

- Collects data



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- Stores rules for using the data
- The ability to reason
- The ability to learn // uses machine learning
- ... by adapting what it does
- ... for example, from mistakes to not make them again // result from previous decisions impacts future
- ... by changing its own rules
- ...by changing its own data
- ...by being trained
- Makes one or more predictions (to make a decision)
- Find/analyse patterns